Fluid Flow Through Fluidized Beds

Final Lab Report
Unit Operations Lab #3
October 14, 2019

Group 1/Tuesday/PM
Marium Murtaza
Adisa Hadzic
Kellen Krupa
Cheryl Slykas

Instructor: Dr. Alan Zdunek

Executive Summary

The objective of this lab is to form fluidized bed, find minimum fluidization velocity from two different materials, fine silica sand and resin beads, record the gas pressure drop of the bed solid particles as a function of gas velocity, and compare the results with the values found with the Ergun equation. [8] Before the calculations, two materials are tested to find the parameters average particle diameter and the void fractions. The two materials differ in particle size so that the sand void fraction is 0.475, while for resin beads is 0.250. This means that the interstitial gaps are higher for sand and the particles should be able to move easier and be fluidized faster. The resin beads in 6-inch bed is observed first with resulting minimum fluidized velocity $\overline{V}_{OM}$ of 0.016 m/s after which the pressure drop stabilizes and the bed reaches fluidization. When observing the bed height variations as a function of flow rate the same pattern is observed where the bed height increases after the point of $\overline{V}_{OM}$. The experimental $\overline{V}_{OM}$ for 10-inch resin beads bed is 0.016 m/s, while $\overline{V}_{OM}$ for sand bed of 6 inch and 10-inch height is 0.025 m/s, and 0.021 m/s, respectively. By using the Ergon equation, the pressure drop per bed height for experimental results below $\overline{V}_{OM}$ for sand (6 inch, 10 inch) are 0.245 kPa/m, and 1.225 kPa/m; for resin beads (6 inch and 10 inch) are 1.959 kPa/m and 0.754 kPa/m, respectively.

In the conclusion, the significant errors are observed in the void fraction calculations mainly due to human error. There is low percent error for pressure drop per unit bed height is observed versus the Ergun equation results, from 0.29% to 2.37%, which proves that the experiment was successful in achieving results similar to the theoretical values.

Introduction

Particles of solids can be fluidized if the fluid is released through the particles bed with velocity large enough to move them. Once set into motion, particles display different behaviors as a function of the velocity.[8] The objective of this experiment is to examine the fluidized bed behavior of two types of bed material, fine silica sand and ion exchange raisin bed, and record the gas pressure drop as a function of gas velocity and compare the results with predicted values found with Ergun equation. When gas is passed through the bed of particles of solids in low velocity, the particles remain immobile while gas flows through the void spaces. The gas flow in the low velocity is flow of the packed bed and it is calculated with the Ergun equation. When velocity is increased, particles become mobile and the bed height increases. The fluidization of bed occurs with higher velocity when particles are moving rapidly, in circular motion, and the bed height is uneven and turbulent. At this point gas flows as “stiff bubbles” that moves the solids particles in the same manner as gas moves in liquids. The pressure drop of fluidized bed does not show significant increase due to system disabling the coefficient of the friction that is present in static bed.

This experiment showcases important uses of fluidized bed in the industries where there is need of overcoming the gravity forces and high gas flow of materials of different densities and different particle size distribution. [4] Fluidized bed industrial applications are in the coal gasification, petrochemical industry catalytic processes, powder mixing devices, food, detergent and cement industries.[4]

The experimental apparatus consists of a 4" plexiglass tube that includes enlarged tube on top to prevent the particle loss. The air is supplied from the main school pressurized air flow through the bottom of the column, and through the porous metal disc that is also a bed support. The air flow is controlled with the needle valve and the air valve that are connected to two rotameters, while the pressure drop is observed with the manometer in the middle of the apparatus. The silica sand and the resin beads are both tested for height of 6 and 10 inches. Air is released slowly, and the pressure drop is observed while is flow is increased in periodic intervals. When the pressure drop remains constant, the bed has reached the fluidized state. At the same time particle size distribution is determined for both materials with the set of sieves, to calculate mass fraction of particle sizes and the average particle diameters that are used in further calculations.

Theory

Fluidized bed reactors are commonly used in chemical engineering to carry out multiphase chemical reactions. The reactor vessel is filled with a solid, fine, granular material usually a catalyst. A gas or fluid is then introduced into the vessel at a certain velocity and it flows through the void spaces between the fine particles. A packed bed reactor occurs at low flow rates, where the gas or liquid flows through the void spaces without the fine particles moving. The particles remain densely packed together due to forces of gravity. The low flow velocity of the gas or fluid does not exert enough force to change the height of a packed bed and the void fraction remains relatively constant. Packed beds experience poor mixing. [5] See Figure 1.

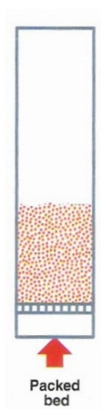

Figure 1: Packed bed [6]

When the velocity of the gas or fluid begins to increase, the force on the particles begin to increase thus causing an increase of the void fraction and bed height. A fluidized bed is reached when the upward velocity of the gas or liquid into the vessel is enough that is begins to break up the bed, increasing the void fraction. This critical velocity is called the minimum fluidization velocity.[7] At this point, the reactor bed particles begin to expand and move randomly, behaving much like a fluid. This allows for complete mixing of the bed particles with the gas or liquid, allowing for a more uniform product and temperature gradient. [5] The height of the bed becomes less defined. "At fluidization conditions, a large fraction of the gas flow passes through the bed in the form of "bubbles." These bubbles are (surprisingly) 'stiff,' and they provide the driving force for the mixing and circulation of the solids." [8] See Figure 2.

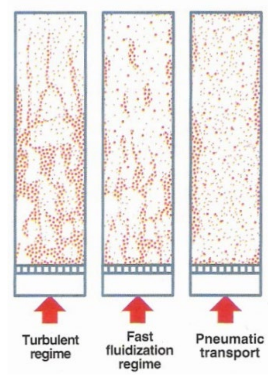

Figure 2: Fluidized beds [6]

As a gas or fluid flows through a packed bed, it experiences a pressure drop due to forces such as friction. The relationship between pressure drop and superficial velocity of air or liquid is described in Figure 3. At zero superficial velocity, the pressure drop in the vessel is zero, and as superficial velocity increases, the pressure drop increases linearly with no change in bed height indicating a fixed, packed bed. At point B on Figure 3, the pressure drop begins to level off and does not change even with increasing velocity, however the bed height begins to increase. This indicates fluidization of the bed and is indicated by points B through D. If, at point D, superficial velocity is decreased again towards zero, a decrease in pressure drop is noted with a constant bed height starting a point C though point E. The bed height does not return to its original packed bed state and the pressure drop does not take its original path because after fluidization, the particles do not settle as tightly packed as when originally put into the vessel.

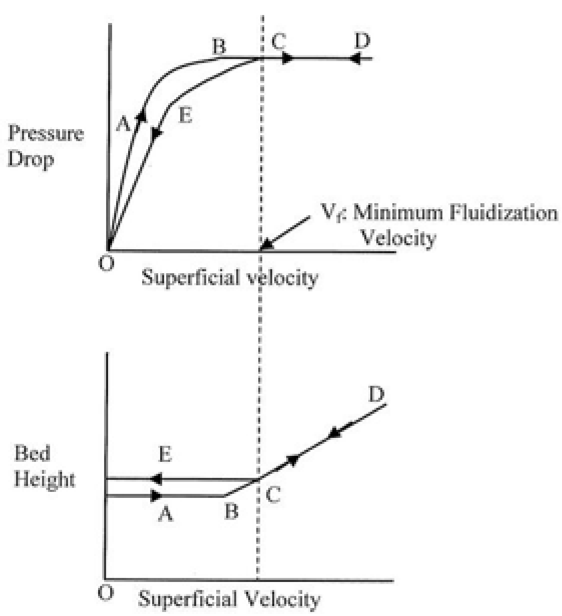

Figure 3: Superficial velocity vs bed height and pressure drop [8]

This pressure drop through a packed bed can be calculated using the Ergun equation,[8]

$$\frac{\Delta P}{L} = 150\frac{\bar{\mu}\mathrm{Vo}(1-\varepsilon)^2}{\Phi_s^2 D_p^2 \varepsilon^3} + 1.75\frac{\rho V_o^2(1-\varepsilon)}{\Phi_s D_p \varepsilon^3} \qquad \text{Equation 1}$$

Where

ΔP = pressure drop across the bed

L = is the bed height

ρ = fluid density

V_o = superficial velocity

ε = void fraction at minimum fluidization

μ = fluid viscosity

Φ_s = sphericity

D_p = particle diameter

The Ergun equation accounts for the pressure loss for both laminar and turbulent flows, the first portion of the equation describing the laminar flow and the second portion describing the turbulent flow. [9] "The laminar region of the equation is independent of the fluid density and has a linear relationship with the superficial velocity. Under turbulent flow conditions the second component of the Ergun equation dominates. Here the pressure drop increases with the square of the superficial velocity and has a linear dependence on the density of the fluid passing through the bed." [9]

At the point where minimum fluidization velocity occurs, the drag force exerted on the bed is equal to the force of gravity on the bed and can be described using Equation 2 below. [8]

$$\frac{\Delta P}{L} = g(1 - \varepsilon_M)(\rho_p - \rho) \qquad \text{Equation 2}$$

Where

ΔP = pressure drop across the bedρ

L = bed volume at minimum fluidization

ε_M = bed voidage at minimum fluidization

ρ_p = density of bed particles

ρ = density of fluid

Combining Equations 1 and 2 gives Equation 3.[8]

$$150\,\frac{\bar{\mu}V_{oM}(1-\varepsilon_M)^2}{\Phi_s^2 D_p^2 \varepsilon_M^3} + 1.75\,\frac{\rho V_{oM}^2}{\Phi_s D_p \varepsilon_M^3} = g(\rho_p - \rho) \qquad \text{Equation 3}$$

The laminar flow term of the Ergun equation is only considered when very small particles are used for the bed. The Reynolds number of the particles is < 1. This yields the equation to estimate minimum fluidization velocity in Equation 4.[8]

$$V_{oM} \approx \frac{g(\rho_p - \rho)}{150\mu}\,\frac{\varepsilon_M^3}{1-\varepsilon_M}\,\Phi_s^2 D_p^2 \qquad \text{Equation 4}$$

Results

In this fluid flow through fluidized beds experiment, the sand and resin beads each were tested using a bed height of 6-inches and 10-inches. The amount of pressure drop across the bed was recorded along with any changes in bed height in the vessel as flowrate varied. In order to begin doing any further calculations, the void fraction and particle diameters were obtained for the sand particles and resin beads. Shown below are the results for those two parameters.

	Average Particle Diameter (Micrometers)	Void Fraction
Sand	193.62	0.475
Resin Beads	476.82	0.25

Table 1: Calculated Values of Average Particle Diameter (Dpavg.) and Void Fraction (ε)

Looking at the calculated results from average particle diameter and void fraction, there is a noticeable difference in size disparity between the two. The sand has an average particle diameter that is much lower than the average particle diameter of the resin beads. The larger the diameter of the resin beads leads to a lower void fraction which shows that is has fewer interstitial gaps. Whereas, the sand particles noticeably have a larger number as their void fraction. This trend shows that the interstitial gaps were more than the ones in the resin beads. The average particle diameter of the sand varied as different particle diameters were determined from the sand and there was a variation of particles of sand when obtaining the void fraction.

After the two parameters above were calculated, the results of the pressure drop per unit bed height (which needed to be calculated in right units) vs. flowrate were obtained for each material with their respective 6-inch and 10-inch bed heights. Figure 4 below displays the two variables of the resin bead material with a bed height of 6 inches. The blue and orange lines show that flowrate was increasing and decreasing respectively during those points. The measured pressure drops per unit bed height as a function of the gas velocity (or flowrate in our case) for both increasing and decreasing flowrates can be utilized to identify where the minimum fluidization velocity is located [8]. In figure 3, the pressure drop vs. superficial velocity was obtained and the minimum fluidization velocity is located where the increasing and decreasing flowrates both meet. Looking at this trend, the blue vertical line in figure 4 appears to be the location of the minimum fluidization velocity, $\overline{V}_{OM}$, which is at a flowrate of 106.67 L/min. After the minimum fluidization velocity, the increasing and decreasing flowrates come close to a constant value which shows the pressure drop per unit area is not varying as much. At the point after the minimum fluidization velocity, the bed of particles doesn't move much and are constant.

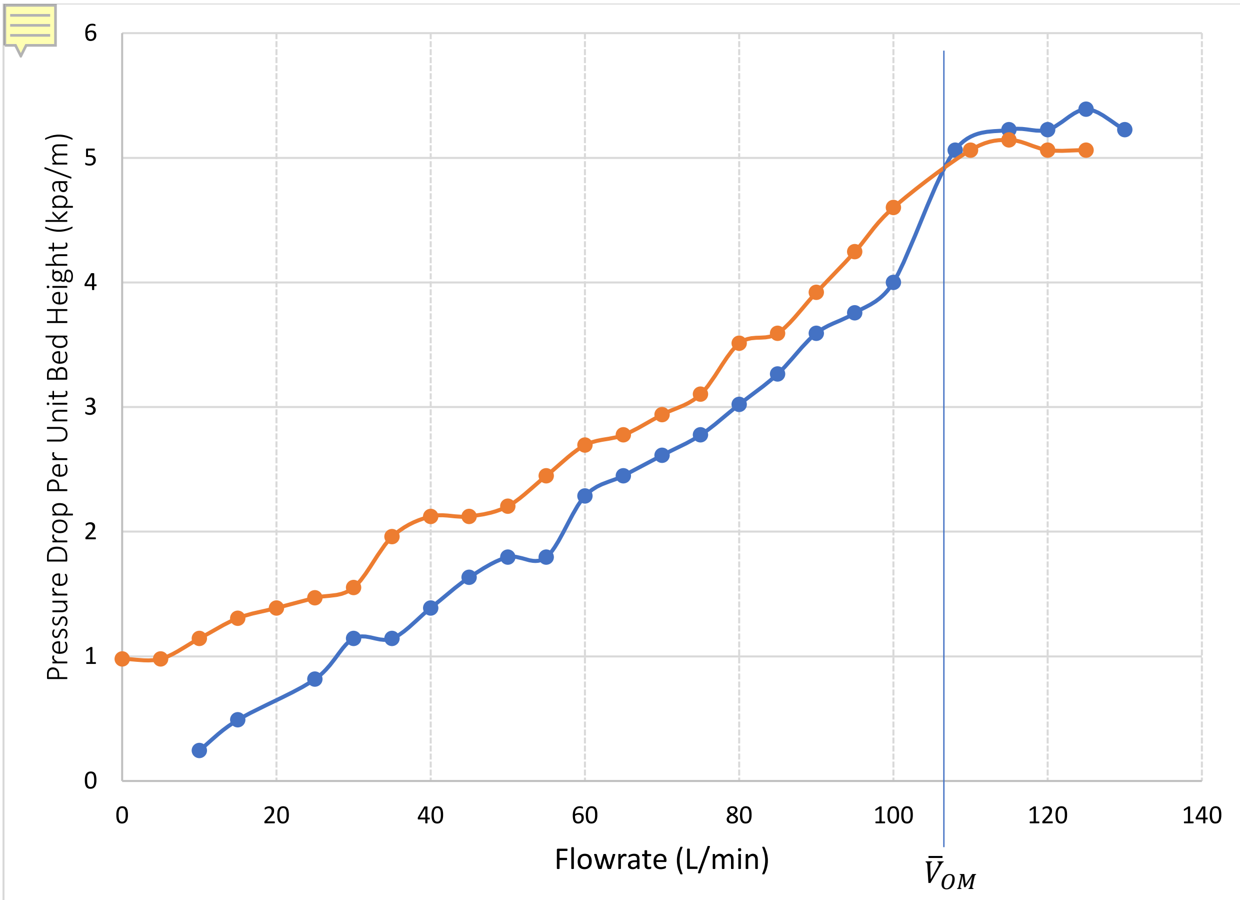

Figure 4: Pressure Drop per Unit Bed Height vs. Flowrate for 6 inch Resin Beads (Blue line is for Increasing Flowrate, Orange line is for Decreasing Flowrate)

Next, the bed height vs. flowrate is plotted in figure 5 below. The blue lines from point to point describe increasing flowrates and orange line is for decreasing flowrates. Comparing the bed height vs. superficial velocity (or flowrate) in figure 3, we can see the same pattern. The location where the increasing and decreasing flowrates just begin to have a similar increasing slope is the point where the minimum fluidization velocity $\overline{V}_{OM}$ is located as shown with the blue vertical line in figure 5. This point, not surprisingly, also occurs at the same minimum fluidization velocity found in figure 4.

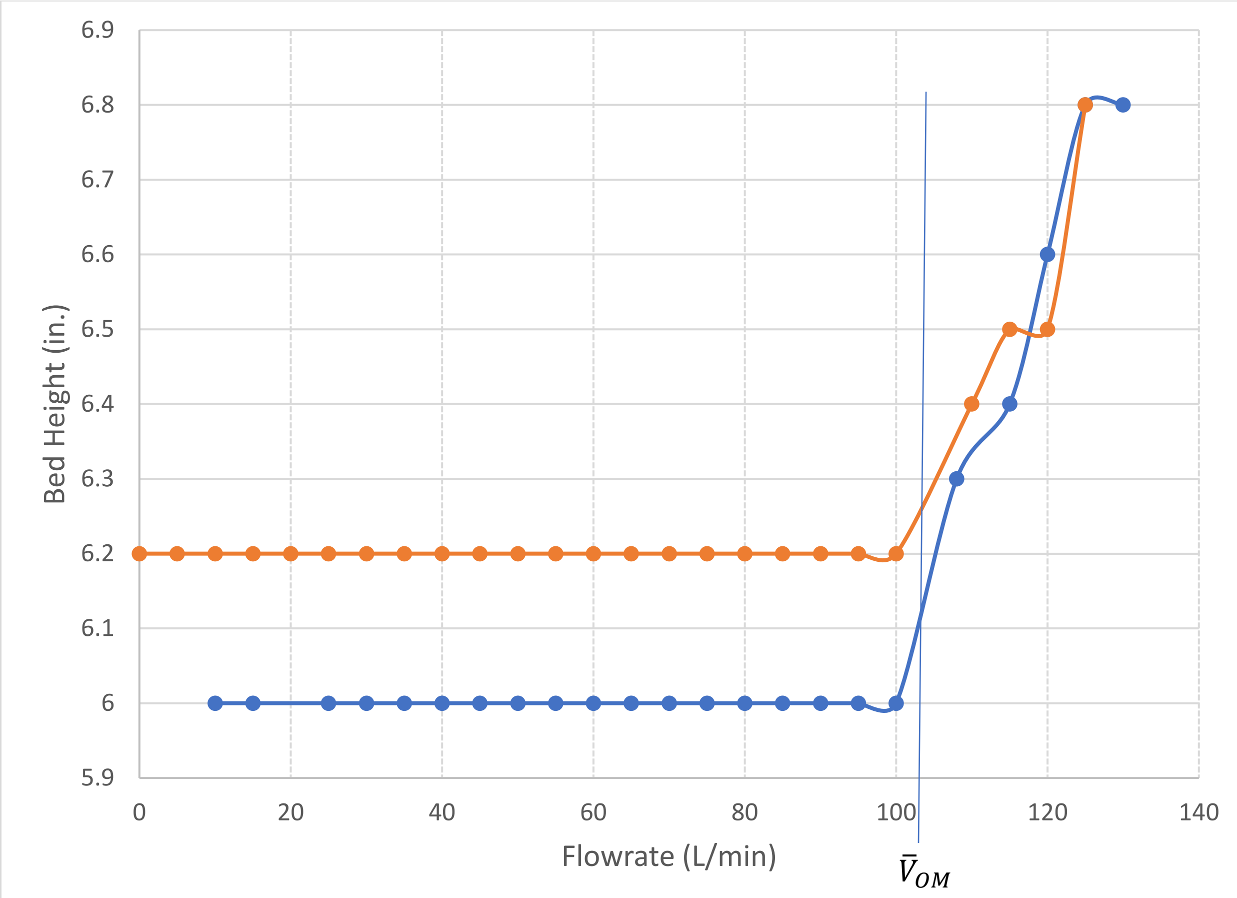

Figure 5: Bed Height vs. Flowrate (Gas Velocity) for 6 inch Resin Beads (Blue is for Increasing Flowrate, Orange is for Decreasing Flowrate)

This typical pattern is also the same for other materials and bed heights as shown in appendix I. After the location is found in these graphs, the minimum fluidization velocity is calculated. The experimental minimum fluidization velocity is by knowing the pressure drop per unit bed height at which the minimum fluidization velocity is located by the graph. Then, using Ergun's equation (equation 1), the left-hand side was known, and the right hand side was filled in with the according values presented in appendix I except for Vo. The Vo term was found using goal seek and the values are shown below. Next, the theoretical Vo was found using equation 4. These numbers are shown in table 2 below. By observing the numbers, the theoretical minimum fluidization velocity seems to be much higher than the experimental.

		Minimum Fluidization Velocity (m/s)	
Material	Bed Height	Experimental Vom (m/s)	Theoretical Vom (m/s)
Sand	6-Inch	0.025	0.033
	10-Inch	0.021	0.033
Resin Beads	6-Inch	0.016	0.020
	10-Inch	0.016	0.020

Table 2: Experimental vs. Theoretical Minimum Fluidization Velocities

The pressure drop was found experimentally and converted as shown in appendix III. The theoretical values are found using the Vo values at that pressure and other values shown in tables in appendix I. The trend in values from experimental to theoretical are closely related as shown in table 3. The values below the minimum fluidization velocity do not differ as much.

		Pressure Drop per Unit Bed Height Below Vom (kpa/m)	
		Experimental	Theoretical
Sand	6-inch	0.245	0.244
	10-inch	1.225	1.233
Resin Beads	6-inch	1.959	1.951
	10-inch	0.754	0.737

Table 3: Experimental vs Theoretical Pressure Drop per unit Bed Height

Discussion

Comparison of the measured results with the calculated results reveals errors in predicting the minimum fluidization velocity, $\bar{V}_{OM}$, but consistency in predicting the pressure drop with the Ergun equation. Experimental $\bar{V}_{OM}$ values in Table 2 are much less than the theoretical values calculated using equation 4. These differences in velocities could be due to faulty LPM flowrates when obtaining the pressure drop data. All experimental values were obtained by the human eye, which may have observed incorrect measurements. Also, the literature values for air viscosity and density used in equation 4 may have caused a small error in the theoretical values. The literature values were at ambient temperature but not at the fluidized pressure of the bed.

Conversely, pressure drop values in Table 3 show similar results between the experimental and theoretical values. As shown in Figure 4, the pressure drop levels when the $\bar{V}_{OM}$ is reached. This means the bed of particles is fluidized and large fractions of air in the form of bubbles are able to pass through the bed. Below this point, the Ergun equation is valid when modeling the pressure drop. Above this point, fluidization has been reached because the increase and decrease in flowrate is relatively constant. At the $\bar{V}_{OM}$, the bed particles are fluidized [8].

Figure 5 shows a simultaneous increase in flowrate with increase in bed height. When the flowrate begins to increase, the bed height rises because there is more force on the particles. This is the point at which fluidization is reached.

By observing table 1, the data clearly shows that silica sand has much more interstitial gaps than resin beads as air can pass through more easily than through resin beads. The higher void fraction for sand means it is closer to a particle with open space and there are more gaps between the particles. Resin is closer to the void fraction of a sold particle and has higher sphericity, meaning it is smoother than the sand. Figure 6 below verifies this due to the higher variance in the size distribution for sand particles than for the resin beads. The experimental $\bar{V}_{OM}$ values conflict the particle distribution and void fractions. Results for sand should show a lower velocity than for the resin because there is less resistance for the air to move the sand particles. A lower void fraction requires that the air have a greater velocity to overcome the force of gravity on the particles.

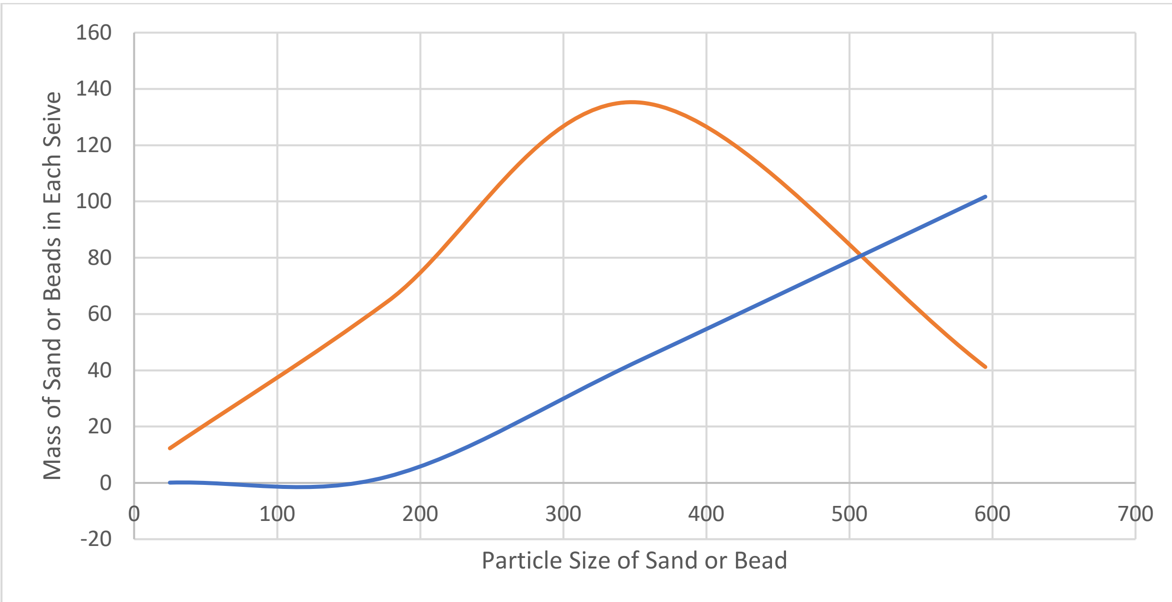

Figure 6: Particle Size Distribution of Sand (orange) and Resin Beads (blue).

The measured void fraction can be used in the Ergun equation because it is the only valid way to measure it in the lab. However, the void fraction was measured without fluid flow. The bed height increases as the minimum fluidization is approached and then reached, which means that the bed voidage will be larger than the one that you measured. Due to this factor, there was a larger percent error in the $\overline{V}_{OM}$ measurements.

The experimentally measured pressure drop per unit bed height can be validated from the theoretical values calculated using the Ergun equation. The measured data agree with the predicted values because those values in Table 3 were obtained for values below the minimum fluidization velocity. This is due to the theory that for very small particles only the laminar flow term of the Ergun equation is significant when the Reynold's number for the particles is less than one. Thus, at low velocities the pressure drop through a packed bed can be determined by the Ergon equation [8].

References

1. Pye, Willard. "Volume 13." Density Determinations of Particles and Sand Grains, edited by Margaret Pye, 1st ed., vol. 13, Department of Geology, 1943, pp. 28–34.

2. J. Davidson, R. Clift, and D. Harrison, Eds., *Fluidization*, 2 nd ed. Academic Press, 1995

3. Engineers Edge, LLC. "Viscosity of Air, Dynamic and Kinematic." Engineers Edge - Engineering, Design and Manufacturing Solutions, 11 June 2018, www.engineersedge.com/physics/viscosity_of_air_dynamic_and_kinematic_14483.htm.

4. "Fluidized Bed." Fluidized Bed - an Overview | ScienceDirect Topics, https://www.sciencedirect.com/topics/chemistry/fluidized-bed.

5. "Fluidized Bed Reactor." https://en.wikipedia.org/wiki/Fluidized_bed_reactor.

6. "Visual Encyclopedia of Chemical Engineering." University of Michigan College of Engineering. http://encyclopedia.che.engin.umich.edu/Pages/Reactors/FBR/FBR.html.

7. "Fluidized Bed." Thermopedia. http://www.thermopedia.com/content/46/.

8. "Fluid Flow Trough Fluidized Beds." CHE 381: Unit Operations Laboratory. University of Illinois at Chicago, 2019.

9. "Pressure Drop Through a Packed Bed." Neutrium. https://neutrium.net/fluid_flow/pressure-drop-through-a-packed-bed/.

Appendix I: Data Tabulation and Graphs

	Particle Density (kg/m^3)	Sphericity
Sand	1538	0.89
Resin Beads	1196	1

Table 4: Determined Values for Particle Density and Sphericity [1] & [2]

	Fluid Density (kg/m^3)	Fluid Viscosity (Pa*s)
Air at 25oC	1.204	1.825*10^-5

Table 5: Determined Values for Fluid Density and Fluid Viscosity of Air at Ambient Temperature [3]

Height (in)	Pressure Drop per Unit Bed Height (kpa/m)	Flow Rate(L/min)
6	4.000	100
6.3	5.062	108
6.4	5.225	115
6.6	5.225	120
6.8	5.388	125
6.8	5.225	130
6.8	5.062	125

Table 6: Bed Heights, Pressure Drops per Unit Bed Height, and Flow Rates of 6 inch Resin Beads

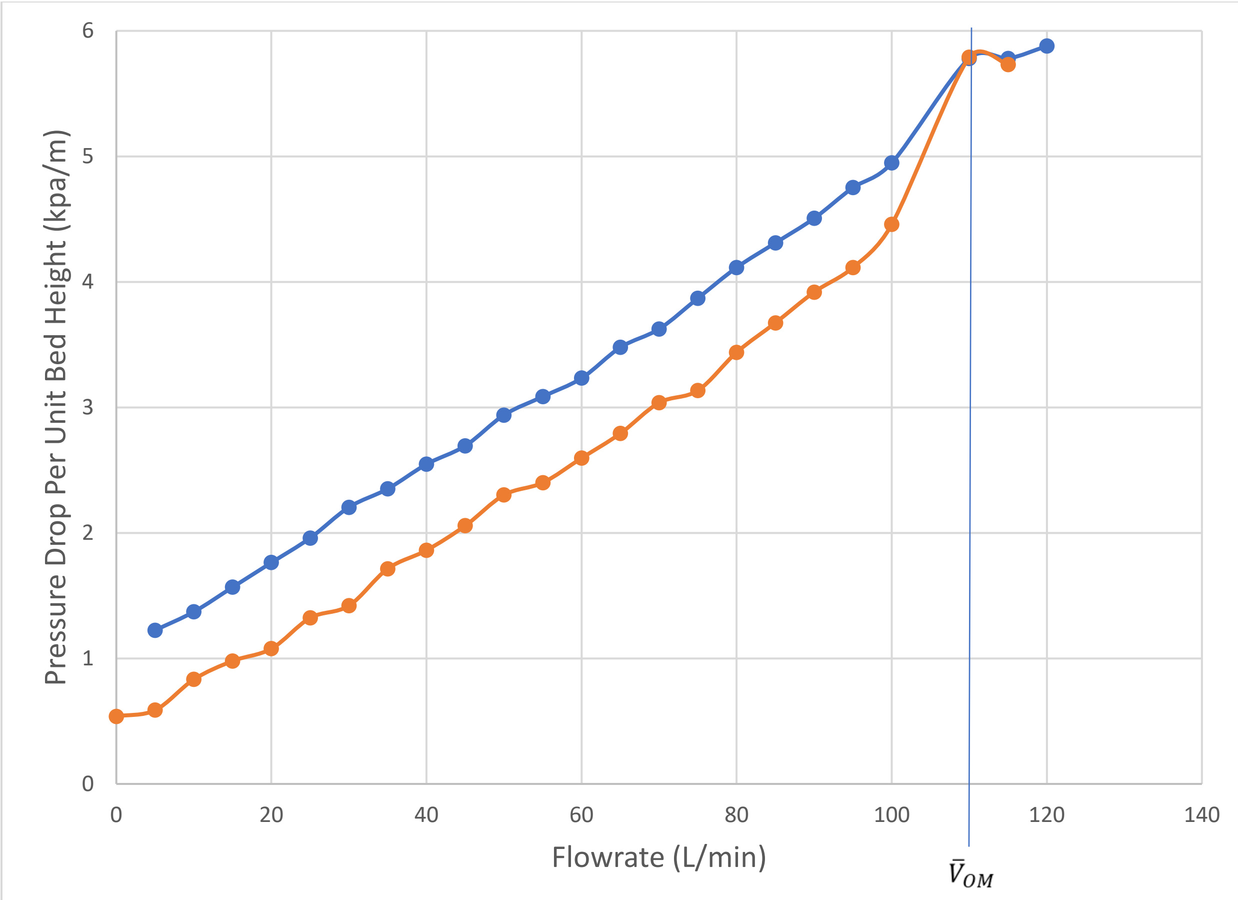

Figure 7: Pressure Drop per Unit Bed Height vs. Flowrate (Gas Velocity) for 10 inch Resin Beads (Blue is for Increasing Flowrate, Orange is for Decreasing Flowrate), $\bar{V}_{OM}$ *at 110 L/min*

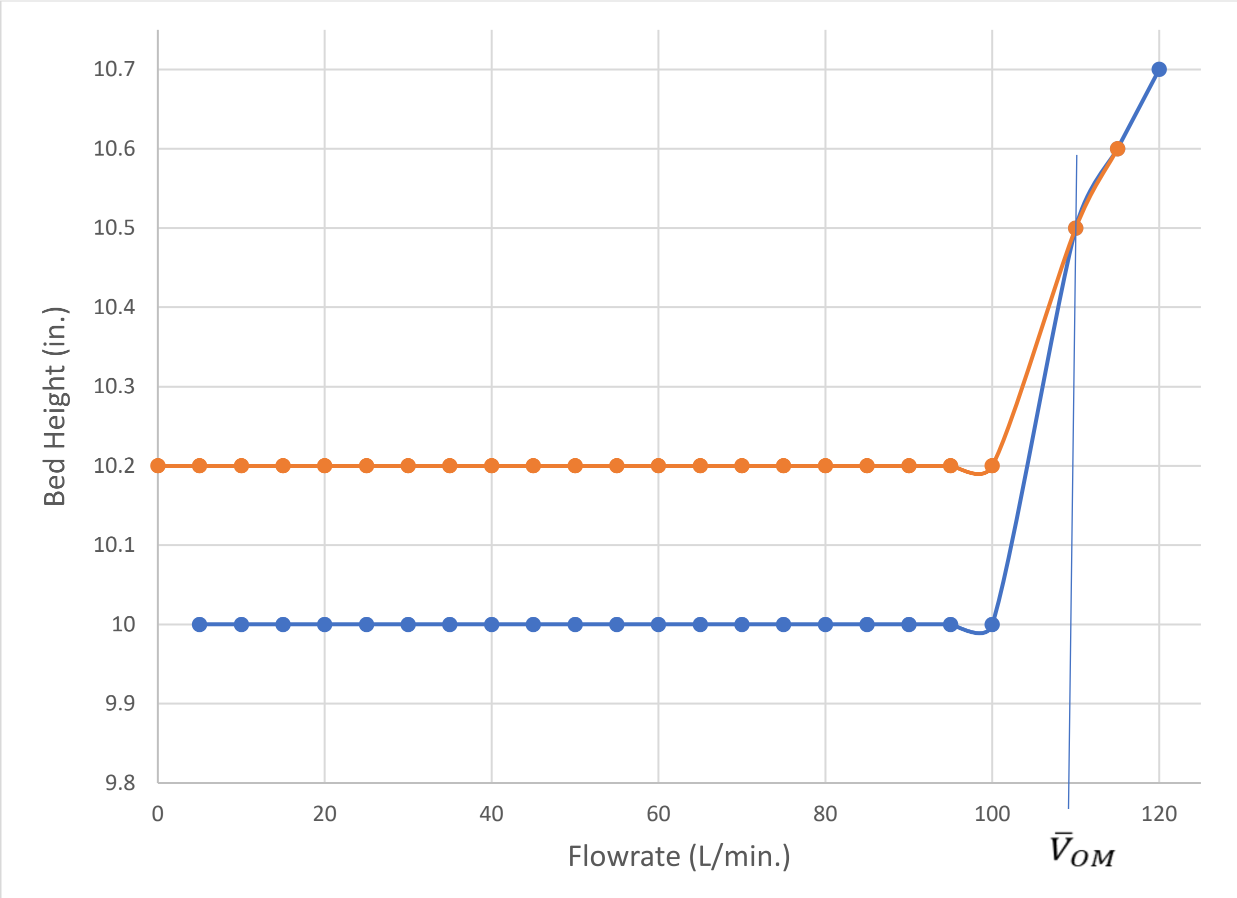

Figure 8: Bed Height vs. Flowrate (Gas Velocity) for 10 inch Resin Beads (Blue is for Increasing Flowrate, Orange is for Decreasing Flowrate $\bar{V}_{OM}$ at 110 L/min

Height (in)	Pressure Drop per Unit Bed Height (kpa/m)	Flow Rate(L/min)
10	4.751	95
10	4.947	100
10.5	5.780	110
10.6	5.780	115
10.7	5.878	120
10.6	5.731	115
10.5	5.789	110
10.2	4.458	100

Table 7: Bed Heights, Pressure Drops per Unit Bed Height, and Flow Rates of 10 inch Resin Beads

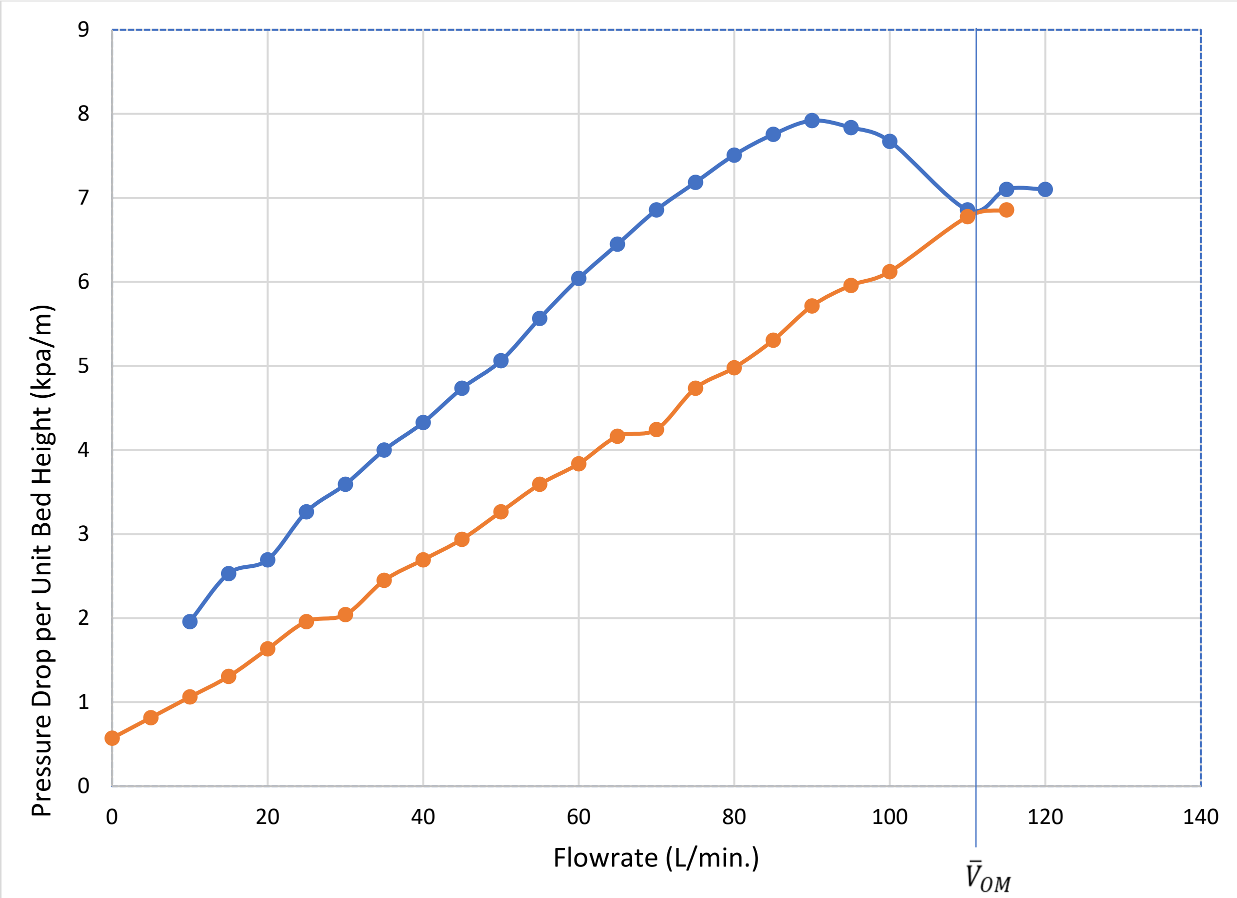

Figure 9: Pressure Drop per Unit Bed Height vs. Flowrate (Gas Velocity) for 6 inch Sand (Blue is for Increasing Flowrate, Orange is for Decreasing Flowrate); $\bar{V}_{OM}$ *at 110 L/min*

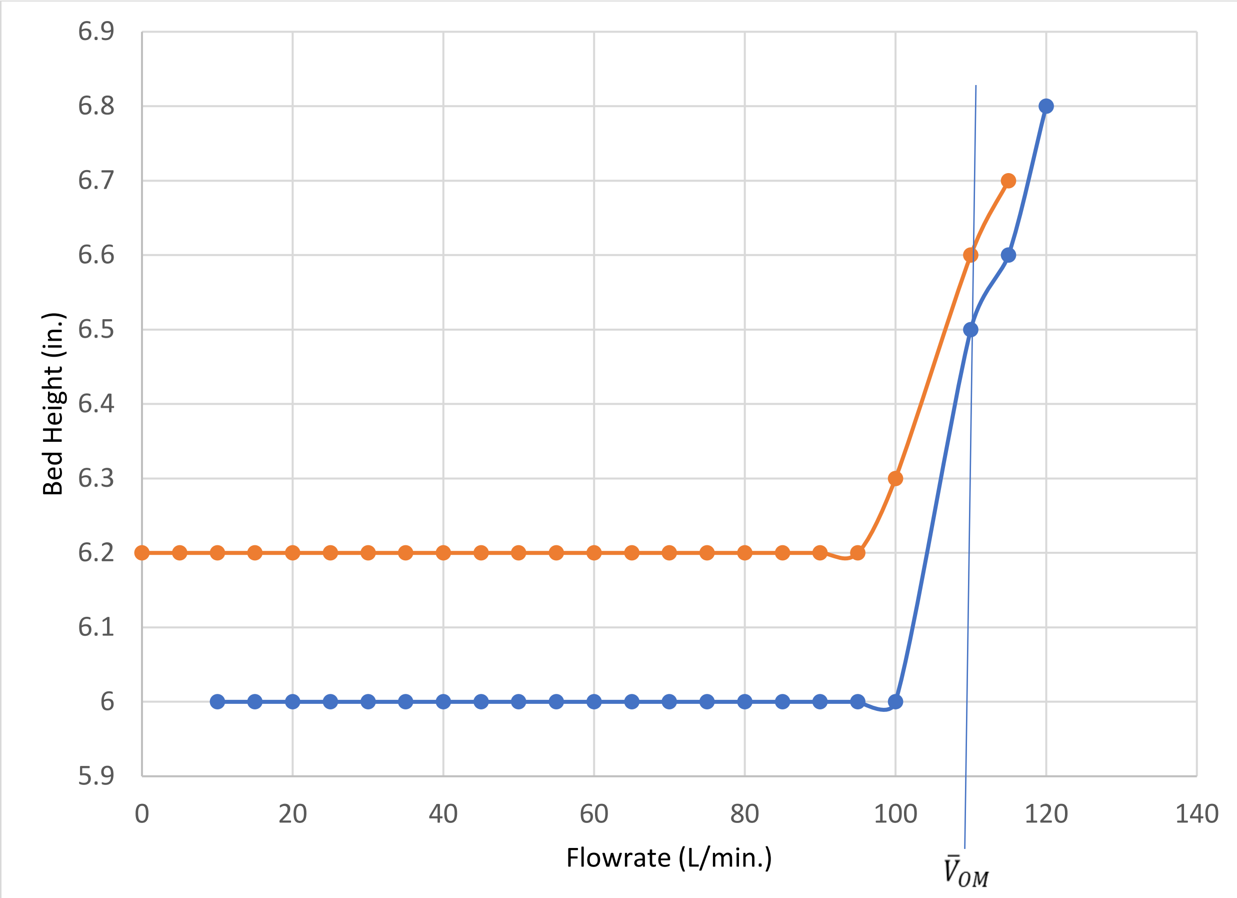

Figure 10: Bed Height vs. Flowrate (Gas Velocity) for 6 inch Sand (Blue is for Increasing Flowrate, Orange is for Decreasing Flowrate); $\bar{V}_{OM}$ at 110 L/min

Height (in)	Pressure Drop per Unit Bed Height (kpa/m)	Flow Rate(L/min)
6.5	6.858	110
6.6	7.103	115
6.8	7.103	120
6.7	6.858	115
6.6	6.776	110
6.3	6.123	100
6.2	5.960	95
6.2	5.715	90

Table 8: Bed Heights, Pressure Drops per Unit Bed Height, and Flow Rates of 6 inch Sand Particles

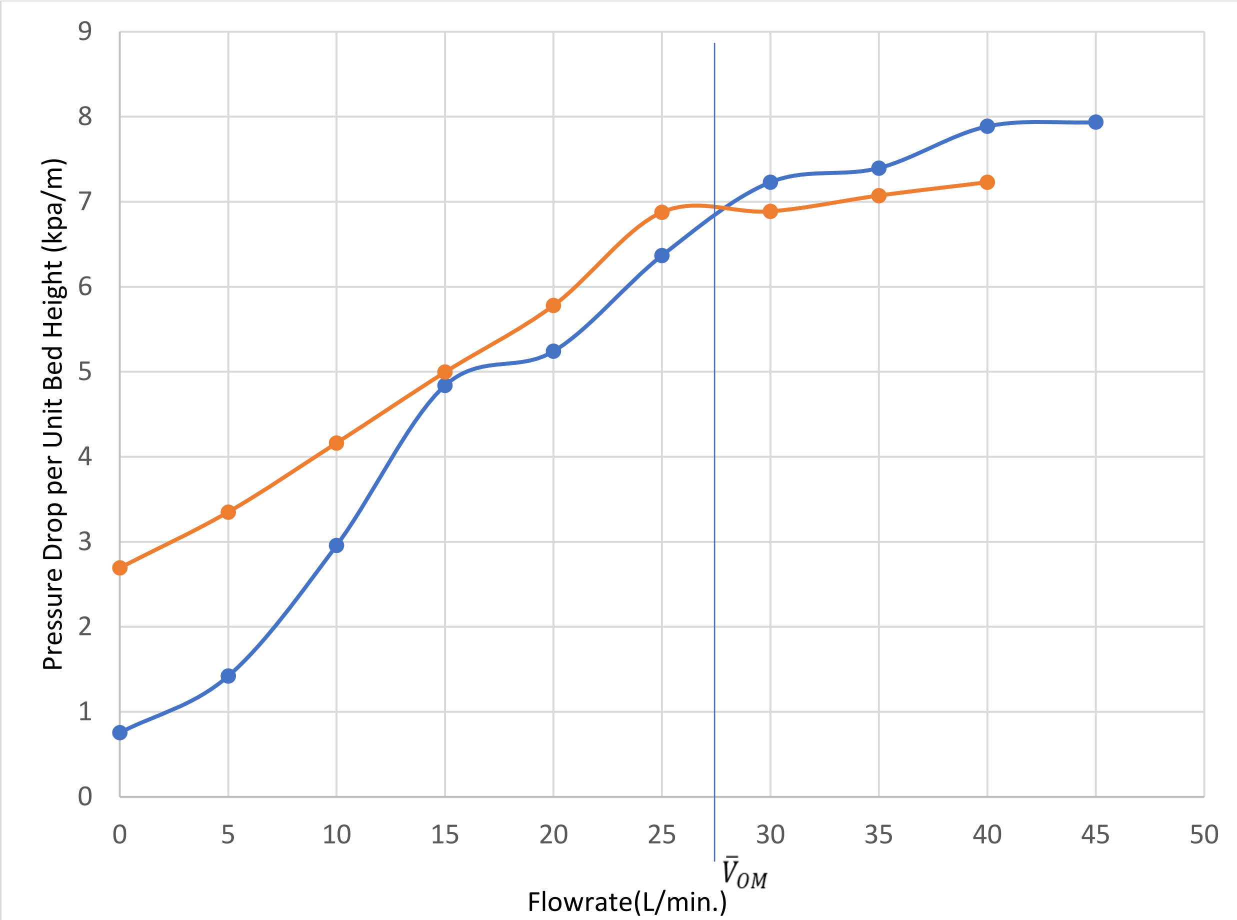

Figure 11: Pressure Drop per Unit Bed Height vs. Flowrate (Gas Velocity) for 10 inch Sand (Blue is for Increasing Flowrate, Orange is for Decreasing Flowrate); $\bar{V}_{OM}$ *at 27 L/min*

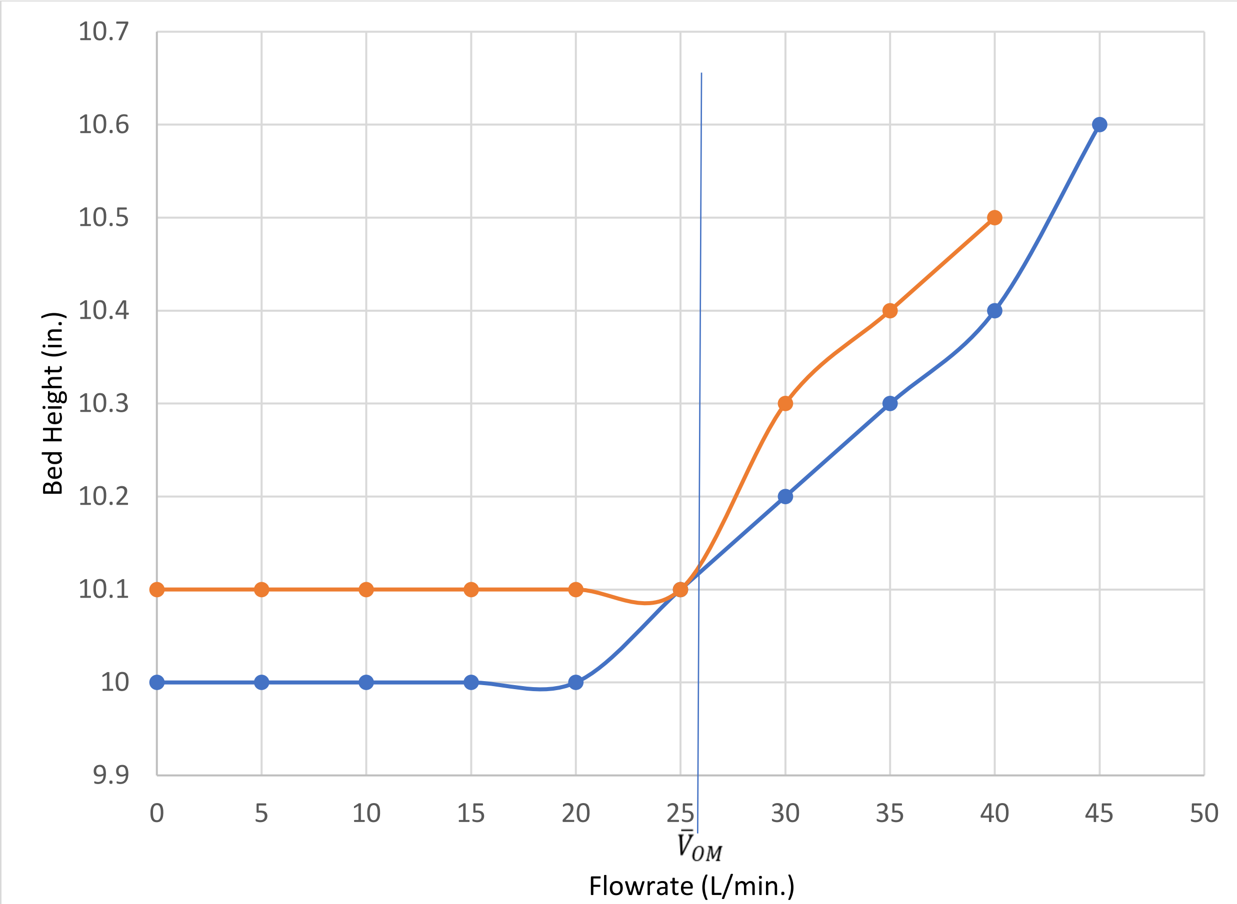

Figure 12: Bed Height vs. Flowrate (Gas Velocity) for 10 inch Sand (Blue is for Increasing Flowrate, Orange is for Decreasing Flowrate); $\bar{V}_{OM}$ at 27 L/min

Height (in)	Pressure Drop per Unit Bed Height (kpa/m)	Flow Rate(L/min)
10	5.241304478	20
10.1	6.36794002	25
10.2	7.23006113	30
10.3	7.396607254	35
10.4	7.886448794	40
10.6	7.935432948	45
10.5	7.23006113	40
10.4	7.073311838	35

Table 9: Bed Heights, Pressure Drops per Unit Bed Height, and Flow Rates of 10 inch Sand Particles

Appendix II: Error Analysis

In the percent error calculations below we can clearly see that the experimental and theoretical values are quite off. This can be due from the multiple human errors encountered during the experimental lab. When increasing and decreasing flowrate the 36.09 LPM flowrate was used and then when we reached the max, we used 426.9 LPM rotameter. This change back and forth could have caused some errors because different flowrates were being increased and decreased leading to a difficult set up in setting a smooth transition for the measurements of pressure drop using the rotameters. Also, it was hard to see the bed height measurement to make sure it was exactly at 6 and 10 inches respectively. The human error in reading the pressure drop may have also occurred. The sphericity for sand was assumed by literature values along with airs density and viscosity at ambient temperature and normal pressure. The void fraction that was determined was not as accurate since some particles were stuck in the sieves and it all depended on how much we shake the sieves for the particles to move across the sieves. This resulted in some human error.

$$Sand\ 6-inch\ \frac{0.033-0.025}{0.033}*100\%=24\%$$

$$Sand\ 10-inch\ \frac{0.033-0.021}{0.033}*100\%=36\%$$

$$Resin\ Beads\ 6-inch\ \frac{0.020-0.016}{0.020}*100\%=22\%$$

$$Resin\ Beads\ 10-inch\ \frac{0.020-0.016}{0.020}*100\%=23\%$$

Looking at the percent errors for pressure drop per unit bed height we can see that the error is not as large when comparing the pressure drop per unit bed height, we obtained experimentally vs. the one using Ergun's equation. This agrees with the fact that there is laminar flow and Ergun equation is sufficient in providing the pressure drops when below minimum fluidization velocity.

$$Sand\ 6-inch\ \frac{0.244-0.0245}{0.244}*100\%=0.29\%$$

$$Sand\ 10-inch\ \frac{1.23-1.225}{1.23}*100\%=0.67\%$$

$$Resin\ Beads\ 6-inch\ \frac{1.95-1.959}{1.95}*100\%=0.45\%$$

$$Resin\ Beads\ 10-inch\ \frac{0.737-0.754}{0.737}*100\%=2.37\%$$

Appendix III: Sample Calculations

Calculation of average particle diameter (for sand):

$$D_{paverage} = \frac{1}{\sum\left\{\frac{xi}{D_{pi}}\right\}} \quad [8]$$

$$D_{paverage} = \frac{1}{\left(\frac{0.16}{595}\right) + \left(\frac{0.53}{355}\right) + \left(\frac{0.25}{177}\right) + \left(\frac{0.05}{25}\right)}$$

$$D_{paverage} = \frac{1}{0.0052}\ \mu m^{-1}$$

$$D_{paverage} = 193.62\ \ \mu m$$

Calculation of void fraction (for sand):

$$\varepsilon = \frac{Amount\ of\ sand\ in\ beaker}{Amount\ of\ water\ needed\ to\ fill\ interstitial\ gaps}$$

$$\varepsilon = \frac{40mL}{19mL}$$

$$\varepsilon = 0.475$$

Pressure Drop per Unit Bed Height conversion (first pressure drop value of 6 inch resin beads):

$$0.15\ manometer\ (in.of\ water) * \frac{248.84Pa}{1in.} * \frac{1}{6in.} * \frac{1in.}{39.37meters} = 0.158\frac{Pa}{m}$$

Experimental Minimum Fluidization Velocity Calculation (sand, 6-inch):

$$\frac{\Delta P}{L} = \frac{150\ \bar{V_o}\ \mu}{\phi_s^2 D_P^2} * \frac{(1-\varepsilon)^2}{\varepsilon^3} + \frac{1.75\rho\ \bar{V_o}^2}{\phi_s D_P} * \frac{(1-\varepsilon)}{\varepsilon^3} \quad [8]$$

$$5.78\frac{kpa}{m} = \frac{150\ \bar{V_o}\ 1.825 * 10^{-5}}{0.89^2 * 0.00019362^2} * \frac{(1-0.475)^2}{0.475^3} + \frac{1.75 * 1.204\ \bar{V_o}^2}{0.89 * 0.00019362} * \frac{(1-0.475)}{0.475^3}$$

Goal-seeked until left hand side matched right hand side by changing Vo value.

$$5.78\frac{kpa}{m} = 5.78\frac{kpa}{m}\ when\ \bar{V_o} = 0.025\frac{m}{s}$$

Theoretical Minimum Fluidization Velocity Calculation (sand, 6-inch):

$$\bar{V}_{oM} \approx \frac{g(\rho_p - \rho)}{150\mu} * \frac{\varepsilon_m^3}{1-\varepsilon_m} * \phi_s^2 D_P^2 \quad [8]$$

$$\bar{V}_{oM} \approx \frac{9.8(1538 - 1.204)}{150 * 1.825 * 10^{-5}} * \frac{(0.475)^3}{1-0.475} * (0.89)^2 * (0.00019362)^2$$

$$\bar{V}_{oM} \approx 0.033\frac{m}{s}$$

Theoretical Pressure Drop per Unit Bed Height:
Using Ergun's Equation and making sure we match the same Vo that was using in that specific pressure below Vom for each material and bed height. This example is calculating for 6-inch Sand.

$$\frac{\Delta P}{L} = \frac{150\,\overline{V_o}\,\mu}{\phi_s^2 D_P^2} * \frac{(1-\varepsilon)^2}{\varepsilon^3} + \frac{1.75\rho\,\overline{V_o}^2}{\phi_s D_P} * \frac{(1-\varepsilon)}{\varepsilon^3} \quad [8]$$

$$\frac{\Delta P}{L} = \frac{150 * 0.00000103 * 1.825 * 10^{-5}}{(0.89 * 0.00019362)\text{^}2} * \frac{(1-0.475)^2}{0.475^3} + \frac{1.75 * 1.204 * 0.00000103^2}{0.89 * 0.00019362} * \frac{(1-0.475)}{0.475^3}$$

$$\frac{\Delta P}{L} = 0.244\,\frac{kpa}{m}$$

Appendix IV: Individual Team Contributions

NAME: Kellen Krupa

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	Present on both days
Pre-Lab Editing	1	
Experimental Objective (Prelab)		
Apparatus (prelab)	2	Research and creating graphic
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	1	Putting sections together and reviewing edit
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)	6	Research, reviewing results, and writing
Conclusion (final lab)		
References (final and/or prelab)	0.5	Typed citations and sources used
Data Tabulation/Graphs (final)		

Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	2	
Total	20.5	

NAME: Cheryl Slykas

	TIME (HOURS)	**DESCRIPTION**
Operator (Both Lab Days)	8	Present both lab periods
Pre-Lab Editing	1	
Experimental Objective (Prelab)	0.5	
Apparatus (prelab)		
Materials & Supplies (prelab)	1	
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	1	
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)	4	Research and writing
Results (final lab)		
Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)	2	Particle size distribution and void fractions
Error Analysis (final lab)		
Sample Calculations (final Lab)	1	
Power Point Presentation	1	
Total	20.5	

NAME: Adisa Hadzic

	TIME (HOURS)	**DESCRIPTION**
Operator (Both Lab Days)	8	
Pre-Lab Editing	2	
Experimental Objective (Prelab)		
Apparatus (prelab)	2	

Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	1	
Executive Summary (final lab)	5	
Introduction (Final lab)	3	
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)		
Conclusion (final lab)	2	
References (final and/or prelab)	0.5	
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	3	
Total	25.5	

NAME: Marium Murtaza

	TIME (HOURS)	**DESCRIPTION**
Operator (Both Lab Days)	8	Accomplished pre-lab and lab experiment.
Pre-Lab Editing	2	Editing of overall lab
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)	2	Procedure of lab
Project Safety Evaluation (prelab)		
Final Lab Editing	2	Final corrections to lab
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	13	Results and making sure calculations are correct
Discussion (final lab)	0	

Conclusion (final lab)	0	
References (final and/or prelab)	1.5	
Data Tabulation/Graphs (final)	5	Data/Graphs input
Error Analysis (final lab)	2.5	
Sample Calculations (final Lab)	3	
Power Point Presentation	0	
Total	39	